

Business Requirements Document

i Draft / Pre-baseline. This BRD translates the direction in Vision and Strategy into business and functional requirements using the controlled terminology in the Glossary and Data Dictionary. Both upstream documents remain draft, and this alignment does not constitute approval, MoSCoW Pass #1 sign-off, baselining, UAT acceptance or release authorisation.

1. Purpose

This BRD defines the intended pre-production P5 product while preserving what is already implemented, what is only designed, what is deferred, and what remains an assumption requiring stakeholder validation. It does not imply that retrospective requirements were approved before development.

P5 is a governed RCM intelligence layer for UAE health-insurance operations. A user asks a question; the agent selects typed tools; deterministic code queries governed views, computes compliance verdicts, or produces chart specifications; the grounding validator blocks unsupported figures; role checks constrain access; and human approval remains mandatory for any outbound appeal.

The [Vision and Strategy](#) page is the upstream source for the product north star, strategic outcomes, scope boundaries, delivery direction, rollout approach and business case. The [Glossary and Data Dictionary](#) page controls draft terminology, current synthetic field definitions, semantic metrics and unresolved data-definition questions. Where an approved future baseline or change record supersedes either draft, the approved record will govern.

Strategic alignment and document hierarchy

Artifact	Role in the lifecycle	Relationship to this BRD
Vision and Strategy	Defines product direction and strategic intent.	Upstream source; currently Draft v0.1 and not approved.
Business Requirements Document	Defines business needs, scope and controlled requirements.	Current document; Draft / Pre-baseline.

Glossary and Data Dictionary	Defines controlled terminology, synthetic schemas, metrics and open data questions.	Supporting BRD artifact; Draft / Pre-baseline.
Stakeholder Register and RACI	Defines proposed roles, accountability and decision authority.	Supporting BRD artifact; assignments require validation.
As-Is and To-Be Processes	Defines current assumptions and intended operating processes.	Supporting BRD artifact; process validation remains pending.
Compliance, Privacy and Security Requirements	Defines proposed business obligations and required compliance evidence.	Supporting BRD artifact; not legal advice or approved compliance interpretation.
Requirements Traceability Matrix	Connects requirements to evidence, design and UAT.	Continuously maintained Governance Spine.
RAID and Decision Log	Controls unresolved risks, assumptions, issues, dependencies and decisions.	Continuously maintained Governance Spine.
Change Control and Sign-Off	Controls approved changes and separates baseline, UAT and release decisions.	Continuously maintained Governance Spine.
NFR , UI and UAT artifacts	Provide provisional downstream requirement and acceptance inputs.	Must be reconciled during the SRS and UAT stages.

2. Business problem

Provider and TPA teams currently depend on fragmented extracts, dashboards, manual denial triage, repeated analyst requests, and manual appeal preparation. The portfolio evidence establishes synthetic baselines including a 33.6% claims-denial rate and material denied/recoverable values, but those figures are demonstration evidence rather than an operating provider's production baseline.

The desired product should reduce time to insight, prioritise recoverable denials before filing windows close, make every number traceable, support compliance review, and give executives, analysts, and auditors role-appropriate views without permitting the language model to invent figures or make regulated decisions.

3. Business objectives

ID	Objective	Measure at baseline	Evidence status
BO-01	Give executives self-service, governed RCM insight	Adoption target to be set	Assumption requiring validation
BO-02	Prioritise recoverable denials before the resubmission window closes	Operational target to be set	Assumption requiring validation
BO-03	Reduce analyst effort for common RCM questions	Response-time and workload targets to be set	Assumption requiring validation
BO-04	Prepare evidence-backed appeal drafts while retaining named-human approval	Approval and cycle-time targets to be set	Assumption requiring validation
BO-05	Make every displayed figure and decision-support artefact traceable	Zero unsupported numbers is an implemented design principle	Implemented for agent responses; UI not implemented
BO-06	Support repeatable	Config-driven onboarding gate	Implemented

	acceptance of new governed data extracts		
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Strategic outcome trace

Strategic outcome	BRD objective alignment	Principal requirement trace	Current interpretation
SO-01 — Governed executive self-service	BO-01	BR-02; FR-07; FR-15; FR-17	Backend aggregate controls partly exist; Executive Insight remains designed/not implemented.
SO-02 — Recoverable-denial prioritisation	BO-02	BR-03; FR-03; FR-16	Backend recoverability ranking is implemented; complete operational UI workflow is not.
SO-03 — Reduced analyst reconciliation and appeal-preparation effort	BO-03 and BO-04	BR-03; BR-04; FR-01; FR-08; FR-10; FR-11; FR-12	Mixed implementation: governed backend functions exist, while the workbench and complete appeal workflow do not.
SO-04 — Compliance and audit reconstruction	BO-05	BR-05; FR-09; FR-12; NFR-05; SEC-07	Grounding and some version metadata exist; persistent audit and evidence presentation remain incomplete.
SO-05 — Safe AI decision support	BO-05	BR-04; BR-05; FR-02; FR-04; FR-05; FR-10; FR-11; FR-14; SEC-08; SEC-09	Core backend safety controls exist; named-human application workflow remains incomplete.

SO-06 — Repeatable governed data acceptance	BO-06	BR-06; FR-13; FR-14; NFR-06	Implemented Gen2 capability.
SO-07 — Usable, accessible and supportable pre-production product	Cross-cutting across BO-01–BO-06	FR-07–FR-18; NFR-03–NFR-16; SEC-01–SEC-10; UAT-01–UAT-12	Intended baseline outcome; substantial product and readiness work remains.
SO-08 — Progress from answering to noticing	Future strategic outcome; no approved baseline objective	FR-19–FR-22	Gen3 capability; designed/not implemented and separately gated.

Strategic trace does not change requirement priority or delivery status. The detailed requirement records remain authoritative for those attributes.

4. Stakeholders and users

- Executive Sponsor — business outcome and funding accountability.
- Product Owner — scope, priority, and acceptance accountability.
- Provider RCM/AR Operations Owner — process and operational acceptance.
- TPA Operations Lead — payer, denial, and appeal operating model.
- RCM Analyst — investigation, denial queue, appeal draft review.
- Executive Viewer — aggregate KPIs and governed insights only.
- Compliance and Regulatory Owner — regulatory interpretation and approval.
- Information Security and Privacy Owner — security, residency, and access controls.
- Data Owner/Steward — extract, mappings, quality, and semantic definitions.
- Platform/Support Owner — environment, monitoring, incident, backup, and recovery.
- UAT Lead and business testers — acceptance execution and evidence.

All role assignments and approval authorities require validation; no personal names are assumed.

5. Scope

In scope for the intended pre-production baseline

- Governed agent orchestration over approved synthetic/provider extracts.

- Typed tools for governed queries, deterministic compliance checks, recoverable-denial ranking, chart specifications, model metadata, and appeal drafting.
- Strict grounding validation and failed-closed handling of unsupported numbers.
- Role-based access for executive/viewer, analyst, and admin/auditor personas.
- Executive Insight, Analyst Workbench, Audit Console, KPI, queue, evidence, appeal, insights, and profile screens defined by the approved UI pack.
- Human review and approval of appeal drafts; no autonomous sending.
- Config-driven dataset portability, version stamping, truth regeneration, rollback, and dry-run evaluation gate.
- Traceability from requirement to design, evidence, and UAT scenario.

Out of scope / deferred

- Production EHR, payer-portal, clearinghouse, payment, or member-contact write integrations.
- Autonomous appeal submission or any other action that changes an external system.
- Gen3 analytical tools (`compare_periods` , `decompose_change` , `detect_anomalies` , findings feed) until separately approved and implemented.
- Gen3 adaptive intake inference, including automated table-role/grain/semantic mapping.
- Gen3 Tier 2 multi-table relationship discovery.
- Gen4 autonomous workers and Track X analytical upgrades.
- Model retraining and production cohort scoring pipelines.

6. Controlled status taxonomy

Label	Meaning
Implemented	Present in current source and supported by tests or run evidence
Designed/not implemented	Defined by approved architecture/UI/decision material but absent from current source
Deferred	Explicitly assigned to a future phase or excluded from this baseline
Assumption requiring validation	Validation/evidence label only; not a delivery status. Proposed business, operational, security, or compliance position requiring validation.

7. Current verified state

Implemented

- Agent loop, six registered tool contracts, system prompt, response assembler, result store, and strict numeric grounding validator.
- Deterministic rules engine and compliance checks.
- Read-only semantic layer and governed query metrics.
- Real recoverable-denial composition and ranking.
- Chart-spec validation bound to an existing query result.
- Code-enforced viewer/analyst/admin role boundaries.
- Evaluation harness, 55-item golden set, judge integration, and historical run logs.
- Gen2 configuration spine, artifact generators, portable golden truths, one-command onboarding, rollback, and dataset-version/config-hash stamping.
- 245 automated tests passing in an isolated verification run on 20 August 2026; 55/55 dry-run validation passed.

Designed/not implemented

- FastAPI/API gateway and the nine approved UI screens.
- Persistent application audit log and evidence-inspector UI.
- Full appeal-letter content, approval endpoint, export, and submission workflow.
- Production `score_member` serving and model explanations.
- Production authentication, SSO, IAM integration, notifications, and support tooling.
- Gen3 adaptive intake and analytical reasoning capabilities.

i Status interpretation: Gen2 completion means that configuration-driven portability, truth regeneration, onboarding and rollback are implemented. It does not mean that the complete three-surface pre-production product, API, authentication, persistent audit service, appeal workflow, operational environment or Gen3 capability is implemented.

Repository HEAD and test evidence are the implementation source of truth. Historical July status documents are superseded where later repository close-out evidence exists. Design or specification wording alone does not prove implementation.

Deferred

- Autonomous outbound actions, Gen4 workers, Track X methods, scheduled scoring, multi-table intake, production write integrations, and model retraining.

8. Prioritisation — MoSCoW Pass #1 (on BRD requirements)

These priorities remain proposed MoSCoW Pass #1 priorities until stakeholder sign-off; no approval has been recorded. Priority does not mean implementation status: a Must may still be Designed/not implemented.

Priority	Requirement IDs	Rationale
Must	BR-01–BR-06, FR-01–FR-14, NFR-01–NFR-08, NFR-15, SEC-01–SEC-10, UAT-01–UAT-12	Minimum governed pre-production product and safety boundary
Should	FR-15–FR-18, NFR-09–NFR-14, NFR-16	Important operating, adoption, evidence, and readiness capability
Could	FR-19–FR-22	Future analytical/adaptive-intake capability after the baseline is accepted
Won't in this baseline	FR-23–FR-26	Autonomous or production-write capability explicitly deferred

Baseline rule: after approval, any change to priority, requirement text, evidence status, or acceptance criterion requires a change record and impact assessment.

Must Requirement Justification and Ownership Register

Requirement ID	Requirement-specific Must rationale	Proposed accountable owner role	Dependency IDs	Measurable baseline acceptance criterion	Delivery status	Implementation evidence / validation note
BR-01	A confirmed Provider/TPA operating model is necessary to prevent gaps or conflicts in RCM ownership, hand-offs and delegated authority.	Provider RCM/AR Operations Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-07	A stakeholder-validated operating model identifies Provider/TPA hand-offs and delegated authorities, and an executed process walkthrough has no unresolved authority gap.	Designed/not implemented	The provider/TPA operating model and role-defined responsibilities remain an assumption requiring validation. No completed end-to-end acceptance evidence is claimed.
BR-02	Executive self-service must preserve aggregate-only access so insight does not	Product Owner — proposed pending requirements-	DEP-05, DEP-07	Executive/viewer users complete KPI and insight workflows while	Designed/not implemented	The backend viewer boundary is implemented; the complete

	expose claim or member detail.	baseline sign-off		UI/API negative tests block all record-level claim or member data.		executive presentation requirement, including UI, is not implemented.
BR-03	Analysts need a governed end-to-end investigation path to act on denials without reconciling fragmented tools manually.	Provider RCM/AR Operations Owner — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-07	An analyst completes denial, recoverability, compliance and evidence investigation through the UI/API, with displayed results reconciling to governed sources.	Designed/not implemented	Semantic, rules and recoverable-denial backend components exist; the required API/UI investigation workflow is not implemented.
BR-04	Named-human control is the primary safety boundary preventing an AI or tool	Product Owner — proposed pending requirements-	DEP-01, DEP-05, DEP-07	Every appeal or external action remains pending until a named	Designed/not implemented	The human-approval safety principle exists, but the named-

	from authorising regulated external action.	baseline sign-off		authorised human acts, and tests prove no model or tool can approve or send; individual appeal authority remains pending under D-BRD-07.		human workflow is not implemented. Requirement ownership does not assign individual appeal authority; D-BRD-07 remains Pending.
BR-05	End-to-end lineage is required to defend every displayed figure and support audit, challenge and correction.	Product Owner — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-08	Every displayed figure resolves to source, query, parameters, execution time and dataset/config version, while an unsupported figure is blocked.	Designed/not implemented	Backend grounding and dataset/config stamping exist; complete end-to-end traceability and evidence presentation are not implemented.
BR-06	Generation gating prevents	Data Owner/Steward —	DEP-03,	A valid generation is	Implemented	Config-driven onboarding

	invalid data, artefacts or truths from becoming the accepted operating baseline.	proposed pending requirements-baseline sign-off	DEP-08	accepted and logged; an invalid generation is rejected and restores the prior accepted data, configuration, artefacts and truths.		g for configured extracts is implemented and supported by current verification evidence.
FR-01	Typed governed queries prevent uncontrolled SQL and make quantitative answers repeatable and reviewable.	Provider RCM/AR Operations Owner — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-07	Approved quantitative queries execute through typed tools only; invalid parameters and free-text SQL attempts are rejected and the result envelope is stamped.	Implemented	Query parameters validated; result envelope stamped
FR-02	Deterministic versioned	Compliance and	DEP-02,	Known test inputs	Implement	Verdict and rules

	compliance logic is necessary because legal or contractual verdicts cannot depend on generative variation.	Regulatory Owner — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-07	return the expected deterministic verdict and rules version on repeat runs, and an agent override attempt is rejected.	ed	version returned
FR-03	Governed recoverability ranking focuses operational effort on eligible denials before filing windows expire.	Provider RCM/AR Operations Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-03, DEP-05, DEP-07	The queue returns only eligible in-window denials and each ranking and deadline field reconciles to governed amount, filing-window and overturn inputs.	Implemented	Only eligible in-window denials returned
FR-04	Fail-closed numeric grounding is release-	Product Owner — proposed	DEP-03, DEP-05,	A deliberately unsupported	Implemented	Ungrounded figures fail closed

	critical because unsupported figures can misdirect financial and operational decisions.	pending requirements-baseline sign-off	DEP-08	ed numeric statement is not displayed and produces a grounding-blocked trace; supported statements reconcile exactly to trusted results.		
FR-05	Server-enforced role boundaries are required to prevent prompt-based privilege escalation and inappropriate data access.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-05, DEP-07	Role-matrix negative tests prove each persona cannot invoke prohibited tools, routes or data access.	Implemented	Code-enforced role capabilities are implemented in the dispatcher.
FR-06	Binding charts to trusted session results	Product Owner — proposed pending	DEP-05, DEP-07	A whitelisted chart renders only from	Designed/not implemented	The backend chart contract is implemented

	prevents a visualisation from introducing unsupported or mismatched data.	requirements-baseline sign-off		an existing session result; an invalid query ID, field or chart type is rejected and no unsupported value is displayed.	mented	ed; the required rendered UI capability is not implemented.
FR-07	Executive Insight is a minimum user surface for governed aggregate decision support and source transparency.	Product Owner — proposed pending requirements-baseline sign-off	DEP-05, DEP-07	An executive completes KPI, narrative, chart, insight and source workflows at approved viewports; values reconcile and aggregate-only access remains enforced.	Designed/not implemented	UI design evidence exists; implemented Executive Insight source, functional verification and stakeholder acceptance evidence are absent.
FR-08	Analyst Workbench is the minimum operational	Provider RCM/AR Operations	DEP-01, DEP-03,	An analyst completes Ask P5, denial-	Designed/not implemented	UI design evidence exists; implement

	surface for denial investigation, evidence review and appeal preparation.	Owner — proposed pending requirements-baseline sign-off	DEP-05, DEP-07	queue, appeal-review and source workflows, including filter, denied and error states, with governed results.	ment ed	ed Analyst Workbench source, functional verification and stakeholder acceptance evidence are absent.
FR-09	Audit Console is required to investigate actions, evidence, versions and access decisions across the product.	Product Owner — proposed pending requirements-baseline sign-off	DEP-05, DEP-06, DEP-07	An authorised admin searches append-only events and retrieves complete user, time, parameters, hashes, versions, response and approval fields; an alteration attempt is rejected.	Designed/not implemented	Audit Console design evidence exists; the persistent audit service, implemented console and acceptance evidence are absent.
FR-10	Governed draft generation	Product Owner — propose	DEP-01, DEP-	Governed claim evidence	Designed/not	The current appeal-

	reduces manual preparation while preserving human review and prohibiting autonomous approval or send.	d pending requirements-baseline sign-off	02, DEP-03, DEP-05, DEP-07	produces an editable draft marked pending review; evidence and prohibited -language checks pass and no automatic approval or send path exists; individual appeal authority remains pending under D-BRD-07.	implemented	letter body is a stub and the complete workflow is not implemented. Requirement ownership does not assign individual appeal authority; D-BRD-07 remains Pending.
FR-11	A complete immutable approval event is necessary to prove who decided what, when and for which resource.	Product Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-05, DEP-07	An authorised human action records identity, time, resource and decision once; a	Designed/not implemented	The approval event is designed/not implemented. Requirement ownership does not

				duplicate is blocked and the immutable event is retrievable ; the authorised role remains pending under D-BRD-07.		assign individual appeal authority; D-BRD-07 remains Pending.
FR-12	Evidence metadata must be visible end to end so users can reconcile outputs to governed sources and configurations.	Data Owner/Steward — proposed pending requirements-baseline sign-offs	DEP-03, DEP-05, DEP-07	The evidence view resolves source view, query ID, parameters, execution time, config version and hash and reconciles them to the displayed output.	Designed/not implemented	Backend lineage elements are partly implemented; the complete evidence page and end-to-end lineage presentation are not implemented.
FR-13	Atomic onboarding and rollback	Data Owner/Steward —	DEP-03,	A valid extract passes	Implemented	Mutation-swap and

	protect the accepted data generation when an extract, artefact or truth fails a gate.	proposed pending requirements-baseline sign-off	DEP-08	schema, enum and window checks and regenerate s tagged artefacts/truths; an invalid extract restores every prior accepted component and retains the decision log.		rollback tests pass
FR-14	Separating read-only agent access from ingestion permissions limits data-integrity and unauthorised-write risk.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-07	Agent-plane attempts to mutate source or replica data are denied, ingestion credentials are segregated, and retained tests demonstrate the boundary.	Implemented	The read-only agent plane and separated ingestion boundary are implemented in the current architecture and code boundary.

NFR-01	Zero tolerance for unsupported numeric content is the core business-trust and financial-safety control.	Product Owner — proposed pending requirements-baseline sign-off	DEP-05, DEP-08	Across the retained grounding test set, zero unsupported numeric statements are displayed and every blocked case produces a traceable error state.	Implemented	Implemented backend; zero tolerance.
NFR-02	Versioned deterministic processing is required for repeatability, defect diagnosis and defensible comparison across runs.	Engineering/Delivery Lead — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-08	Repeated execution with identical input and component versions returns identical deterministic outputs, and the applicable versions are retained.	Implemented	Implemented backend.

NFR-03	Least privilege enforced outside prompts is essential to protect regulated data and privileged functions.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-05, DEP-07	Server-side negative tests block unauthorised role and prompt-escalation attempts; production IAM, SSO/MFA and role-matrix evidence is retained.	Designed/not implemented	Implemented in dispatcher ; production IAM not implemented.
NFR-04	Aggregate-only executive access minimises privacy exposure while retaining business insight.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-03, DEP-05, DEP-07	Viewer tests return aggregate results only and block claim/member detail across UI, API and export paths.	Designed/not implemented	Implemented backend; UI not implemented.
NFR-05	Append-only searchable audit evidence is required for	Compliance and Regulatory Owner —	DEP-05, DEP-06,	Required audit fields are append-only and	Designed/not implemented	Designed/not implemented as an

	investigation, accountability and compliance review.	proposed pending requirements-baseline sign-off	DEP-07	searchable, and retained negative evidence shows alteration and deletion attempts are rejected.	mented	application service.
NFR-06	Full rollback of rejected generations prevents mixed-version data, configuration and truth states.	Data Owner/Steward — proposed pending requirements-baseline sign-off	DEP-03, DEP-08	For a rejected generation, checksums for data, configuration, artefacts and truths match the prior accepted state after rollback.	Implemented	Implemented and tested.
NFR-07	An approved and tested availability commitment is necessary before users rely on the pre-production	Platform/Support Owner — proposed pending requirements-	DEP-06, DEP-07	An accountability-owner-approved support window and availability target are	Designed/not implemented	Proposed 99.5% during agreed support hours; validate. No completed

	service for governed workflows.	baseline sign-off		recorded before UAT, and monitoring evidence shows the tested service meets that recorded value.		end-to-end acceptance evidence is claimed.
NFR-08	UAE residency control is release-critical because unverified storage or inference routing may breach applicable obligations.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-02, DEP-04, DEP-05, DEP-07	Retained data-flow and routing evidence shows production health-data storage and inference remain in the UAE; any exception is documented and authorised before use.	Designed/not implemented	Architecture intent only; validate legal and model availability. No completed end-to-end acceptance evidence is claimed.
SEC-01	Applicable obligations and	Compliance and Regulatory	DEP-01,	A versioned obligations	Designed/not	Assumption requiring

	controller/processor roles must be known before controls and accountability can be baselined.	ry Owner — proposed pending requirements-baseline sign-off	DEP-02	register and Provider/TPA controller/processor role record identify the applicable jurisdiction, contracts, duties and accountable roles before baseline approval.	implemented	validation. No completed end-to-end acceptance evidence is claimed.
SEC-02	UAE storage and processing controls reduce unlawful cross-border handling of regulated health data.	Compliance and Regulatory Owner — proposed pending requirements-baseline sign-off	DEP-02, DEP-04, DEP-05	Environment and data-flow evidence shows regulated health data is stored and processed in the UAE; any exception has written lawful authority	Designed/not implemented	Requirement proposed from official law; deployment not implemented.

				before use.		
SEC-03	Verified In-Region routing prevents an approved configuration from silently using an unsuitable global or geographic route.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-02, DEP-04, DEP-05	Configuration and build evidence disables unsuitable Global/Geo routing and verifies the approved In-Region UAE route; a negative route test is retained.	Designed/not implemented	Designed/not implemented.
SEC-04	Named identities, MFA/SSO and least privilege are required to authenticate users and enforce accountable access.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-05, DEP-07	Named identities use approved SSO/MFA and server-side least privilege; the role matrix passes and no shared or prompt-derived	Designed/not implemented	Dispatcher checks implemented; production IAM not implemented.

				privilege is accepted.		
SEC-05	Aggregate-only viewer access and controlled detail/export paths minimise unnecessary disclosure of claim and member data.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-03, DEP-05, DEP-07	Viewer tests expose aggregate s only, and detail or export is blocked unless an authorised role and approved purpose are evidenced and audited.	Designed/not implemented	Backend boundary implemented; UI/export controls not implemented.
SEC-06	Encryption and governed key rotation protect regulated data against interception or storage compromise.	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-05, DEP-06, DEP-07	Approved encryption protects data in transit and at rest, key ownership is recorded, and a retained rotation test completes without data loss or	Designed/not implemented	Assumption requiring validation. No completed end-to-end acceptance evidence is claimed.

				unauthoris ed access.		
SEC-07	Protected, synchronised and retained audit records are necessary for reliable investigation and evidence preservation.	Compliance and Regulatory Owner — proposed pending requirements-baseline sign-off	DEP-02, DEP-05, DEP-06, DEP-07	Audit records are append-only, protected, searchable and time-synchronised; retention and legal-hold behaviour is tested against the approved schedule.	Designed/not implemented	Designed/not implemented.
SEC-08	Human decision authority must remain separate from AI decision support for appeals and every external action.	Compliance and Regulatory Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-02, DEP-05, DEP-07	AI output remains decision support, every approval or external action requires a named authorised human, and no autonomous action	Designed/not implemented	The human-control safety principle exists, but the approval workflow is not implemented; individual appeal authority

				route is available.		remains Pending under D-BRD-07.
SEC-09	Restricting determinations to approved deterministic rules prevents unsupported clinical, coverage, eligibility, fraud or misconduct decisions.	Compliance and Regulatory Owner — proposed pending requirements-baseline sign-off	DEP-02, DEP-05, DEP-07, DEP-08	Negative tests show the system does not issue clinical, coverage, eligibility, fraud or misconduct determinations outside approved deterministic rules.	Implemented	Implemented prompt/tool boundary and tests.
SEC-10	Minimisation, masking and purpose-based export controls reduce privacy exposure across roles and environments .	Information Security and Privacy Owner — proposed pending requirements-baseline sign-off	DEP-01, DEP-02, DEP-03, DEP-05, DEP-07	Role- and environment-based tests demonstrate minimisation, pseudonymisation or masking, purpose control and export	Designed/not implemented	Partly designed; validate. No completed end-to-end acceptance evidence is claimed.

				restriction for representative regulated data.		
UAT-01	The Provider/TPA operating model is release-critical because unclear hand-offs or authority would invalidate the business process.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-01, DEP-07	Executed scenario evidence shows ownership, status, escalation and approval match the stakeholder-validated Provider/TPA process, with no unresolved Severity 1 or 2 defect.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-02	A negative viewer-access test is required to prove aggregate-only scope and prevent record-level disclosure.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-01, DEP-05, DEP-07	The request is blocked, an aggregate alternative is offered and no record-level claim	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance

				or member data appears in the UI, API response or audit output.		e record is claimed.
UAT-03	Analyst result and lineage reconciliation proves that operational insight remains grounded in governed sources.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-03, DEP-05, DEP-07	Displayed results, formatting, query ID, source and configuration lineage reconcile exactly to the governed tool result.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-04	Known-claim compliance testing proves deterministic rules and prevents agent override of governed verdicts.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-02, DEP-03, DEP-05, DEP-07	Known claims return the expected deterministic verdict and rules version, and the agent cannot alter or override the verdict.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.

UAT-05	Recoverable-denial queue testing protects filing deadlines and verifies that ranking uses eligible governed inputs.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-01, DEP-03, DEP-05, DEP-07	Every displayed denial is eligible and within the filing window, and ranking and deadline fields reconcile to governed inputs.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-06	Appeal workflow testing is required to prove pending-review status, named-human control and absence of an autonomous send path.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-01, DEP-02, DEP-05, DEP-07	The draft remains pending until a named authorised human acts and no send path exists; individual appeal authority remains pending under D-BRD-07.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-07	A deliberate unsupported-	UAT Lead —	DEP-03,	The unsupported	Designed/not	The scenario

	figure test proves the fail-closed grounding safety boundary.	proposed pending requirements-baseline sign-off	DEP-05, DEP-07, DEP-08	ed figure is not displayed, an accessible grounding -blocked state appears and a trace ID resolves to the blocked output.	ot implemented	is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-08	Evidence and audit retrieval testing proves that actions and generated outputs are traceable after the event.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-05, DEP-06, DEP-07	Question, tool calls, parameters, result hash, rules/model/config versions, response and any approval event are complete and mutually reconciling .	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-09	Valid/invalid onboarding tests prove generation	UAT Lead — proposed	DEP-03, DEP-07,	The valid generation is accepted;	Designed/not implemented	The scenario is defined, but

	acceptance and complete rollback before new data is trusted.	pending requirements-baseline sign-off	DEP-08	the invalid generation is rejected and restores data, configuration, artefacts and truths to the prior accepted state.	ment ed	business UAT has not been executed and no acceptance record is claimed.
UAT-10	Executive navigation testing confirms the aggregate decision-support surface is usable, governed and source-aware.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-05, DEP-07	At the stakeholder-approved visual baseline in force at execution, the executive completes KPI, insight, chart and source workflows with aggregate scope and correct formatting.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-11	Queue and appeal	UAT Lead —	DEP-05,	Search, filters,	Designed/n	The scenario

	interaction testing confirms analysts can complete priority workflows across normal, empty and error states.	proposed pending requirements-baseline sign-off	DEP-07	selection, deadlines, evidence-readiness and empty/error states work without a blocked required action.	not implemented	is defined, but business UAT has not been executed and no acceptance record is claimed.
UAT-12	Administrative version and access-event testing proves configuration and control evidence is complete and immutable.	UAT Lead — proposed pending requirements-baseline sign-off	DEP-05, DEP-06, DEP-07	Rules, model and configuration versions plus access events are complete, searchable and immutable, and an alteration attempt is rejected.	Designed/not implemented	The scenario is defined, but business UAT has not been executed and no acceptance record is claimed.
NFR-15	Accessible operation is required so primary workflows are usable without a mouse and	Product Owner — proposed pending requirements-	DEP-05, DEP-07	No critical accessibility violation is present; all primary workflows can be completed	Designed/not implemented	UI designs exist; no implemented accessibility audit or keyboard

	status or error information is perceivable.	baseline sign-off		using a keyboard; automated and manual accessibility evidence is retained.		walkthrough exists.
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9. Detailed business requirements

ID	Requirement	Priority	Delivery status	Acceptance summary
BR-01	P5 shall support provider RCM operations with TPA participation using role-defined responsibilities.	Must	Designed/not implemented	Implementation evidence / validation note: The provider/TPA operating model and role-defined responsibilities remain an assumption requiring validation. Acceptance: Provider/TPA operating model approved
BR-02	P5 shall present governed aggregate	Must	Designed/not implemented	Implementation evidence / validation note: The

	insight to executives without claim/member detail.			backend viewer boundary is implemented ; the complete executive presentation requirement, including UI, is not implemented . Acceptance: Viewer cannot retrieve record-level data
BR-03	P5 shall support analyst investigation of denials, recoverability , compliance, and evidence.	Must	Designed/not implemented	Implementation evidence / validation note: Semantic, rules and recoverable-denial backend components exist; the required API/UI investigation workflow is not implemented .

				Acceptance: Approved scenarios pass through UI/API
BR-04	P5 shall retain a named-human decision for every appeal approval or external action.	Must	Designed/not implemented	Implementati on evidence / validation note: The human-approval safety principle is implemented ; the named-human approval workflow is not implemented . Acceptance: No tool or model call can approve/sen d
BR-05	P5 shall make generated figures traceable to governed tool results and	Must	Designed/not implemented	Implementati on evidence / validation note: Backend grounding and dataset/confi g stamping

	dataset/config versions.			exist; complete end-to-end traceability and evidence presentation are not implemented . Acceptance: Unsupported figure is blocked; citations resolve
BR-06	P5 shall accept only governed data generations that pass validation and evaluation gates.	Must	Implemented	Implementation evidence / validation note: Config-driven onboarding for configured extracts is implemented and supported by current verification evidence. Acceptance: Accepted/rejected drop is logged and recoverable

10. Detailed functional requirements

ID	Requirement	Priority	Delivery status	Acceptance summary
FR-01	Answer approved quantitative RCM questions through typed semantic views; no free-text SQL.	Must	Implemented	Query parameters validated; result envelope stamped
FR-02	Compute compliance verdicts in versioned deterministic code, not by the LLM.	Must	Implemented	Verdict and rules version returned
FR-03	Rank recoverable denials using governed amount, filing-window and overturn inputs.	Must	Implemented	Only eligible in-window denials returned
FR-04	Validate every numeric statement against trusted tool results before display.	Must	Implemented	Ungrounded figures fail closed
FR-05	Enforce viewer, analyst, and admin capabilities in code.	Must	Implemented	Implementation evidence / validation note: Code-enforced role capabilities are implemented in the dispatcher. Acceptance: Role-negative scenarios pass
FR-06	Render chart specifications only	Must	Designed/not implemented	Implementation evidence /

	against existing session query results and whitelisted chart types.			validation note: The backend chart contract is implemented; the required rendered UI capability is not implemented. Acceptance: Invalid query ID/spec rejected
FR-07	Provide Executive Insight with KPI band, governed narratives, charts, recent insights, and source affordances.	Must	Designed/not implemented	Approved UI acceptance map passed
FR-08	Provide Analyst Workbench with Ask P5, recoverable-denial queue, appeal review, and source access.	Must	Designed/not implemented	Approved UI acceptance map passed
FR-09	Provide Audit Console with activity, evidence, rules/model versions, and trace inspection.	Must	Designed/not implemented	Audit events searchable and complete
FR-10	Generate appeal drafts from governed claim/evidence data while prohibiting automatic approval or send.	Must	Designed/not implemented	Implementation evidence / validation note: The current appeal-letter body is a stub; the complete drafting workflow is not

				implemented. Acceptance: Draft marked pending review; prohibited- language checks pass
FR-11	Record named approver, time, resource, and decision for approval events.	Must	Designed/not implemented	Duplicate approval blocked; event immutable
FR-12	Expose source view, query ID, parameters, execution time, config version, and config hash for generated evidence.	Must	Designed/not implemented	Implementation evidence / validation note: Backend lineage elements are partly implemented; the complete evidence page and end-to-end lineage presentation are not implemented. Acceptance: Evidence page resolves complete lineage
FR-13	Onboard a configured extract by validating schema/enums/window, appending with a generation tag, regenerating	Must	Implemented	Mutation-swap and rollback tests pass

	artefacts/truths, gating, and rolling back on rejection.			
FR-14	Preserve raw/source data as read-only to the agent plane and separate ingestion permissions.	Must	Implemented	Implementation evidence / validation note: The read-only agent plane and separated ingestion boundary are implemented in the current architecture and code boundary. Acceptance: Agent role cannot mutate source/replica
FR-15	Support KPI dashboards, filters, export controls, and approved formatting.	Should	Designed/not implemented	UI and export controls pass UAT
FR-16	Support a full recoverable-denials queue with search, filters, priority, evidence readiness, and deadlines.	Should	Designed/not implemented	Implementation evidence / validation note: The backend queue feed exists; the complete searchable/filterable UI queue is not implemented. Acceptance: Queue fields

				reconcile to source results
FR-17	Support an insights library with severity, impact, evidence, and drill-down.	Should	Designed/not implemented	Implementation evidence / validation note: Included in the intended pre-production baseline as a Should requirement. The All Insights screen and functional insights library are not implemented. Acceptance: Stored findings have resolvable citations
FR-18	Support profile, notification preferences, and explicit role/access details.	Should	Designed/not implemented	Access display matches server permissions
FR-19	Profile unfamiliar single-table RCM extracts and propose table role, grain, and mappings for human confirmation.	Could	Designed/not implemented	Separate Gen3 approval and implementation required
FR-20	Compare periods and decompose changes with server-computed	Could	Designed/not implemented	Implementation evidence / validation note:

	values and hedged narration.			Future Gen3 analytical/proactive capability with Could priority; not part of the minimum baseline. Acceptance: Separate Gen3 approval and implementation required
FR-21	Detect anomalies with versioned deterministic methods and planted-anomaly tests.	Could	Designed/not implemented	Implementation evidence / validation note: Future Gen3 analytical/proactive capability with Could priority; not part of the minimum baseline. Acceptance: Separate Gen3 approval and implementation required
FR-22	File proactive, cited findings under a least-privilege monitor identity.	Could	Designed/not implemented	Implementation evidence / validation note: Future Gen3 analytical/proactive capability with Could priority; not part of the minimum

				baseline. Acceptance: Separate Gen3 approval and implementation required
FR-23	Submit appeals automatically to payers.	Won't	Deferred	Explicitly excluded
FR-24	Write to production claims/EHR systems.	Won't	Deferred	Explicitly excluded
FR-25	Perform automatic multi-table relationship discovery.	Won't	Deferred	Gen3 Tier 2
FR-26	Run autonomous Gen4 workers or scheduled cohort scoring.	Won't	Deferred	Future decision only

11. Business rules

- The LLM never computes, derives, or invents a number.
- Compliance verdicts come from the versioned rules engine.
- Unsupported questions receive an honest refusal or nearest governed alternative.
- Viewer users receive aggregates only; record/member detail is analyst/admin scope.
- An appeal remains `pending_review` until a named authorised human acts.
- No autonomous outbound or production-write path exists in this baseline.
- The agent uses a read-only data plane; ingestion is a separate operations path.
- A rejected data generation must roll back configuration, artefacts, truths, and appended rows.
- Regulatory rules and filing-window logic are not silently changed through dataset configuration.
- UI mock-up figures and identities are illustrative and must not become test truth.

Business and technical terms, physical fields, controlled values and semantic calculations are described in the [Glossary and Data Dictionary](#). Its definitions remain pre-baseline. Regulatory

rules, filing-window logic and approved business meanings must not be silently changed through dataset configuration.

12. Dependencies

- Approved provider/TPA operating model and data-processing responsibilities.
- Named data, compliance, security, product, and UAT owners.
- Approved UAE deployment and model-routing design.
- Approved data extracts, mapping, retention, security, and legal interpretation.
- API/UI implementation, authentication, audit store, environment, and support model.

The open data-definition decisions DD-01 through DD-14 are maintained in the [Glossary and Data Dictionary](#). They elaborate existing Provider/TPA, regulatory, data, security and UAT dependencies; they do not create approved mappings or production meanings. Material unresolved items must be cross-referenced in the [RAID and Decision Log](#) during the Governance Spine alignment pass.

13. Constraints

- Synthetic data only in the current portfolio implementation.
- Static pinned dataset clock; not a live production refresh.
- Current model call path uses an Anthropic API model ID in code/eval evidence; production UAE-region model hosting is not implemented.
- The latest complete live evaluation is not a full post-fix green run; GQ-038 remains unverified after the targeted rerun.
- No implemented web/API application, production authentication, persistent audit service, or outbound integration.

14. Acceptance and exit criteria

Requirements-baseline approval is approval of:

1. Business context and objectives.
2. Scope and exclusions.
3. Business and functional requirements.
4. Delivery-status classifications.
5. MoSCoW Pass #1.
6. Recorded assumptions, dependencies and risks.
7. Named role-based decision authorities.

8. Requirements-baseline approval authorises backlog creation and implementation planning. It does not constitute UAT acceptance or release approval.
9. Vision and Strategy direction has been reviewed and any unresolved conflict with the BRD is recorded.
10. Glossary, metric definitions and material data-definition decisions have been reviewed by the proposed accountable roles.
11. Supporting artifacts and Governance Spine links resolve to the current versions.

If Vision and Strategy, material data definitions, stakeholder authority or required decisions remain unresolved, the BRD remains Draft / Pre-baseline.

15. Approval status

Role	Decision	Name	Date	Comments
Executive Sponsor	Pending			
Product Owner	Pending			
Provider RCM/AR Operations Owner	Pending			
TPA Operations Lead	Pending			
Compliance and Regulatory Owner	Pending			
Information Security/Privacy Owner	Pending			
Data Owner	Pending			
UAT Lead	Pending			

Revision note — 21 August 2026: Added upstream strategy, terminology and data-definition traceability. No requirement ID, requirement wording, priority, delivery status, evidence note, acceptance criterion or approval state was changed.